



NEW ROCK

The rock debris eventually settles in beds of sediment that form on seabeds, on the floors of lakes, or in sand dunes. Dissolved minerals gradually harden around the rock fragments, cementing them together to create layers of new sedimentary rocks such as sandstone and shale.



◀ **PUSHED UP**
The ocean floor grinding beneath the edges of continents crumples the rocks into fold mountains.



UPHEAVAL

The forces of plate tectonics push sedimentary rocks above sea level and buckle them into mountain ranges. As they are pushed up, the rocks are squeezed, hardened, and recrystallized by the processes that create metamorphic slates and schists. Such rocks form many of the world's highest peaks.

SUBDUCTION

While some rocks are pushed up by colliding tectonic plates, others are dragged down into the mantle beneath the crust. Here, they are exposed to extreme pressure and intense heat, which transforms them into metamorphic rocks such as gneiss, migmatite, and eclogite.



◀ **EXTREME STRESS**
This outcrop of gneiss was formed by intense heat and pressure deep beneath the crust.



MELTDOWN

If subduction drags rocks down far enough, they are likely to melt, forming magma. This wells up toward the surface, and either forms igneous intrusions underground or erupts from volcanoes. This new igneous rock starts to break down as soon as it is exposed to the air, and the cycle starts over again.